

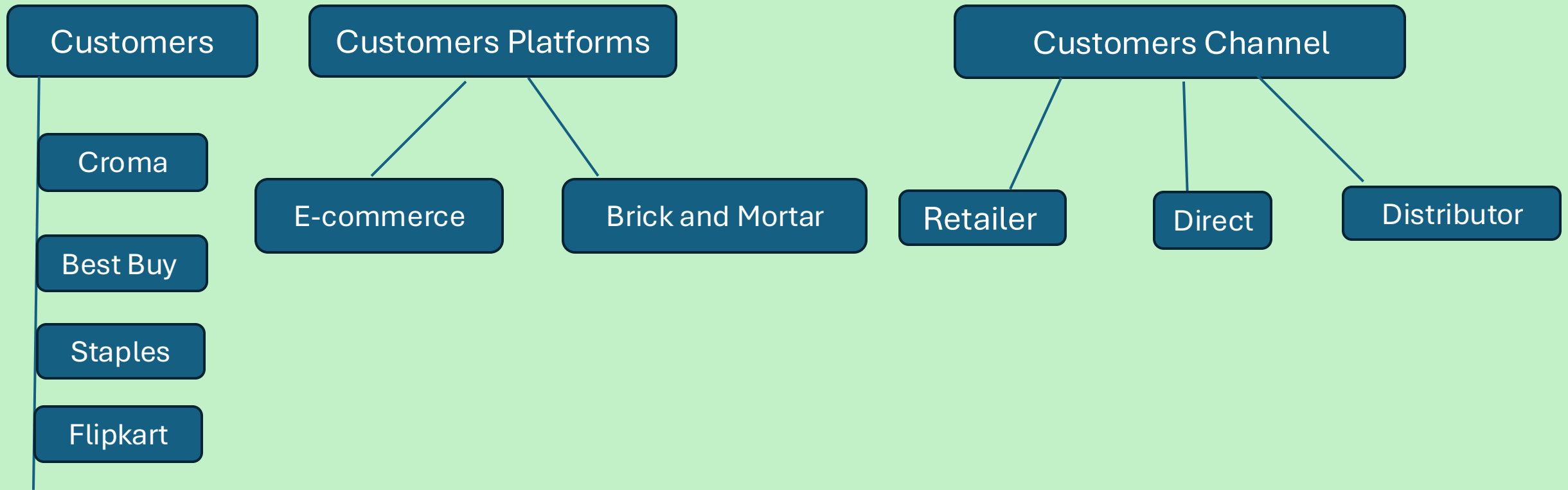
About AtliQ Hardwares and Problem Statement

- - AtliQ Hardwares is a leading hardware company that deals in PCs, printers, mice, and computers and has a global presence.
- - The growing size of Excel files has caused performance issues, making the data process slow and inefficient. To solve this problem, AtliQ Hardwares started a project to collect and analyze the data. MySQL will be used as the database management system to extract useful insights from the data. These insights will help the company improve decision-making and overall business performance.

Project Overview

- - This project is focused on analyzing the provided database and finding useful insights from the data. The database contains information about sales, products, customers, and regions for AtliQ Hardware. The project aims to answer specific questions related to sales reports, market analysis, customer behavior, and supply chain forecasting.

AtliQ Hardwares- Business Model



AtliQ Hardwares- Business Model

Product Division

PC

N&S

P&A

Product Segment

Storage

Peripherals

Notebook

Desktop

Accessories

Croma India Product Wise Sales Report for Fiscal Year -2021

```
SELECT
    s.date, s.product_code,
    p.product, p.variant, s.sold_quantity,
    g.gross_price as Gross_price_per_item,
    g.gross_price*s.sold_quantity as gross_price_total
FROM fact_sales_monthly s
JOIN dim_product p
ON
    p.product_code = s.product_code
JOIN fact_gross_price g
ON
    g.product_code = s.product_code AND
    g.fiscal_year = get_fiscal_year(s.date)
WHERE
    customer_code=90002002 AND
    get_fiscal_year(date)=2021
ORDER BY date asc
Limit 1000000
```

date	product_code	product	variant	sold_quantity	Gross_price_per_item	gross_price_total
2020-09-01	A0118150101	AQ Dracula HDD – 3.5 Inch SATA 6 Gb/s 5400 R...	Standard	202	19.0573	3849.5746
2020-09-01	A0118150102	AQ Dracula HDD – 3.5 Inch SATA 6 Gb/s 5400 R...	Plus	162	21.4565	3475.9530
2020-09-01	A0118150103	AQ Dracula HDD – 3.5 Inch SATA 6 Gb/s 5400 R...	Premium	193	21.7795	4203.4435
2020-09-01	A0118150104	AQ Dracula HDD – 3.5 Inch SATA 6 Gb/s 5400 R...	Premium Plus	146	22.9729	3354.0434
2020-09-01	A0219150201	AQ WereWolf NAS Internal Hard Drive HDD – 8.8...	Standard	149	23.6987	3531.1063
2020-09-01	A0219150202	AQ WereWolf NAS Internal Hard Drive HDD – 8.8...	Plus	107	24.7312	2646.2384
2020-09-01	A0220150203	AQ WereWolf NAS Internal Hard Drive HDD – 8.8...	Premium	123	23.6154	2904.6942
2020-09-01	A0320150301	AQ Zion Saga	Standard	146	23.7223	3463.4558
2020-09-01	A0321150302	AQ Zion Saga	Plus	236	27.1027	6396.2372
2020-09-01	A0321150303	AQ Zion Saga	Premium	137	28.0059	3836.8083
2020-09-01	A0418150103	AQ Mforce Gen X	Standard 3	23	19.5235	449.0405
2020-09-01	A0418150104	AQ Mforce Gen X	Plus 1	82	19.9239	1633.7598
2020-09-01	A0418150105	AQ Mforce Gen X	Plus 2	86	20.0766	1726.5876
2020-09-01	A0418150106	AQ Mforce Gen X	Plus 3	48	19.9365	956.9520
2020-09-01	A0519150201	AQ Mforce Gen Y	Standard 1	138	22.3984	3090.9792
2020-09-01	A0519150202	AQ Mforce Gen Y	Standard 2	72	24.9298	1794.9456

SQL- Query

OUTPUT

Gross Monthly Total Sales Report For Croma

```
SELECT
monthname(s.date) as month,
SUM(ROUND(g.gross_price*s.sold_quantity,2)) as gross_price_total
FROM fact_sales_monthly s
JOIN fact_gross_price g
ON g.product_code = s.product_code and
   g.fiscal_year = get_fiscal_year(s.date)
WHERE
   customer_code = 90002002
GROUP BY s.date
ORDER BY s.date asc;
```

SQL- Query

month	gross_price_total
September	122407.57
October	162687.56
December	245673.84
January	127574.73
February	144799.54
April	130643.92
May	139165.06
June	125735.36
August	125409.90
September	343337.14
October	440562.10
December	653944.72
January	359025.06
February	356607.19
April	379549.74
May	340152.29
June	343702.08

OUTPUT

Yearly Gross Sales Report For Croma India

```
select g.fiscal_year,  
round(sum(s.sold_quantity * g.gross_price)/1000000,2)as "gross_price_total(in mln)"  
from fact_sales_monthly s  
join fact_gross_price g  
on s.product_code = g.product_code  
and get_fiscal_year(s.date) = g.fiscal_year  
where customer_code = 90002002  
group by g.fiscal_year;
```

SQL- Query

fiscal_year	gross_price_total(in mln)
2018	1.32
2019	3.56
2020	6.50
2021	23.22
2022	44.64

OUTPUT

Top Product By Net sales for a financial Year "2021"

```
SELECT
    Product,
    ROUND(SUM(net_sales)/1000000,2) AS net_sales_mln
FROM net_sales n
JOIN dim_customer c
ON n.customer_code=c.customer_code
WHERE fiscal_year=2021
GROUP BY Product
ORDER BY net_sales_mln desc
LIMIT 5;
```

SQL- Query

product	net_sales_mln
AQ BZ Allin1	33.75
AQ Qwerty	27.84
AQ Trigger	26.95
AQ Gen Y	23.58
AQ Maxima	22.32

OUTPUT

Top Market By Net sales for a financial Year "2021"

```
select market,  
round(sum(net_sales)/1000000,2) as net_sales_mln  
from net_sales  
where fiscal_year = 2021  
group by market  
order by net_sales_mln desc  
limit 5;
```

SQL- Query

market	net_sales_mln
India	210.67
USA	132.05
South Korea	64.01
Canada	45.89
United Kingdom	44.73

OUTPUT

Top Customer By Net sales for a financial Year "2021"

```
select c.customer,  
round(sum(net_sales)/1000000,2) as net_sales_mln  
from net_sales s  
join dim_customer c  
using(customer_code)  
where fiscal_year = 2021  
group by customer  
order by net_sales_mln desc  
limit 5;
```

SQL- Query

customer	net_sales_mln
Amazon	109.03
Atliq Exclusive	79.92
Atliq e Store	70.31
Sage	27.07
Flipkart	25.25

OUTPUT

Net Sales % Share By Customers

```
with cte1 as(
  SELECT
    c.customer,
    ROUND(SUM(net_sales)/1000000,2) AS net_sales_mln
    FROM net_sales n
    JOIN dim_customer c
    ON n.customer_code=c.customer_code
  WHERE fiscal_year=2021
  GROUP BY c.customer)
  SELECT
    *,
    net_sales_mln*100/SUM(net_sales_mln) OVER() as pct
  FROM cte1
  ORDER BY net_sales_mln desc
```

SQL- Query

customer	net_sales_mln	pct
Amazon	109.03	13.233402
Atliq Exclusive	79.92	9.700206
Atliq e Store	70.31	8.533803
Sage	27.07	3.285593
Flipkart	25.25	3.064692
Leader	24.52	2.976089
Neptune	21.01	2.550067
Ebay	19.88	2.412914
Electricalsociety	16.25	1.972327
Synthetic	16.10	1.954121
Electricalslytical	15.64	1.898289
Acclaimed Stores	14.32	1.738075
Propel	14.14	1.716228
Novus	12.91	1.566938
Expression	12.90	1.565724
Reliance Digital	12.75	1.547518
walmart	12.63	1.532053

OUTPUT

Net Sales % Share by Region - "APAC"

```
with cte1 as(
SELECT
    c.customer,c.region,
    ROUND(SUM(net_sales)/1000000,2) AS net_sales_mln
    FROM net_sales n
    JOIN dim_customer c
    ON n.customer_code=c.customer_code
WHERE fiscal_year=2021 AND region = "APAC"
GROUP BY c.customer,c.region)
SELECT *,
    net_sales_mln*100/SUM(net_sales_mln) OVER() as pct_share_region
FROM cte1
ORDER BY region,net_sales_mln desc
```

customer	region	net_sales_mln	pct_share_region
Amazon	APAC	57.41	6.967910
Atliq Exclusive	APAC	51.58	6.260317
Atliq e Store	APAC	36.97	4.487086
Leader	APAC	24.52	2.976017
Sage	APAC	22.85	2.773328
Neptune	APAC	21.01	2.550005
Electricalsociety	APAC	16.25	1.972279
Synthetic	APAC	14.14	1.716186
Propel	APAC	14.14	1.716186
Flipkart	APAC	12.96	1.572968

OUTPUT

SQL- Query

Top 3 Products by Total Quantity in Each division

```
with cte1 as (select
    p.division,
    p.product,
    sum(sold_quantity) as total_qty
from fact_sales_monthly s
join dim_product p
    on p.product_code=s.product_code
where fiscal_year =2021
group by p.product),
cte2 as(select
    *,
    dense_rank() over(partition by division order by total_qty desc)as drnk
from cte1)
select * from cte2
where drnk<=3
```

SQL- Query

division	product	total_qty	drnk
N & S	AQ Pen Drive DRC	2034569	1
N & S	AQ Digit SSD	1240149	2
N & S	AQ Clx1	1238683	3
P & A	AQ Gamers Ms	2477098	1
P & A	AQ Maxima Ms	2461991	2
P & A	AQ Master wireless x1 Ms	2448784	3
PC	AQ Digit	135092	1
PC	AQ Gen Y	135031	2
PC	AQ Elite	134431	3

OUTPUT

Supply Chain – Forecast Accuracy

```
with forecast_err_table as(
select
  s.customer_code,
  sum(s.sold_quantity) as total_sold_qty,
  sum(forecast_quantity-sold_quantity) as net_err,
  sum((forecast_quantity-sold_quantity))*100/sum(forecast_quantity) as net_err_pct,
  sum(abs(forecast_quantity-sold_quantity)) as abs_err,
  sum(abs(forecast_quantity-sold_quantity))*100/sum(forecast_quantity) as abs_err_pct
from gdb0041.fact_act_est s
where s.fiscal_year =2021
group by customer_code)
select
e.*,
  c.customer,c.market,
  if(abs_err_pct > 100, 0, 100-abs_err_pct) as forecast_accuracy
from forecast_err_table e
join dim_customer c
using(customer_code)
order by forecast_accuracy desc;
```

SQL- Query

customer_code	total_sold_qty	net_err	net_err_pct	abs_err	abs_err_pct	customer	market	forecast_accuracy
90013120	109547	23985	17.9620	70467	52.7716	Coolblue	Italy	47.2284
70010048	119439	22571	15.8940	75711	53.3139	Atliq e Store	Bangladesh	46.6861
90023027	236189	43773	15.6353	149303	53.3297	Costco	Canada	46.6703
90023026	228988	44504	16.2725	146948	53.7303	Relief	Canada	46.2697
90017051	86823	31244	26.4629	63568	53.8406	Forward Stores	Portugal	46.1594
90017058	86860	23335	21.1761	59473	53.9707	Mbit	Portugal	46.0293
90023028	239081	44242	15.6154	153058	54.0224	walmart	Canada	45.9776
90023024	246397	40836	14.2170	155610	54.1755	Sage	Canada	45.8245
90013124	110898	25218	18.5268	73826	54.2376	Amazon	Italy	45.7624
90015146	147152	63355	30.0964	114189	54.2448	Mbit	Norway	45.7552
90017054	84371	30327	26.4407	62483	54.4761	Flawless Stores	Portugal	45.5239
70027208	33713	13608	28.7568	25784	54.4874	Atliq e Store	Brazil	45.5126
90015147	154897	68970	30.8085	122100	54.5413	Chiptec	Norway	45.4587
80001019	1113979	161269	12.6461	695779	54.5603	Neptune	China	45.4397
90015144	160074	65563	29.0568	123257	54.6262	Sound	Norway	45.3738
00000130	103700	6885	6.7401	60775	54.6630	Logic Stores	Newzealand	45.3370

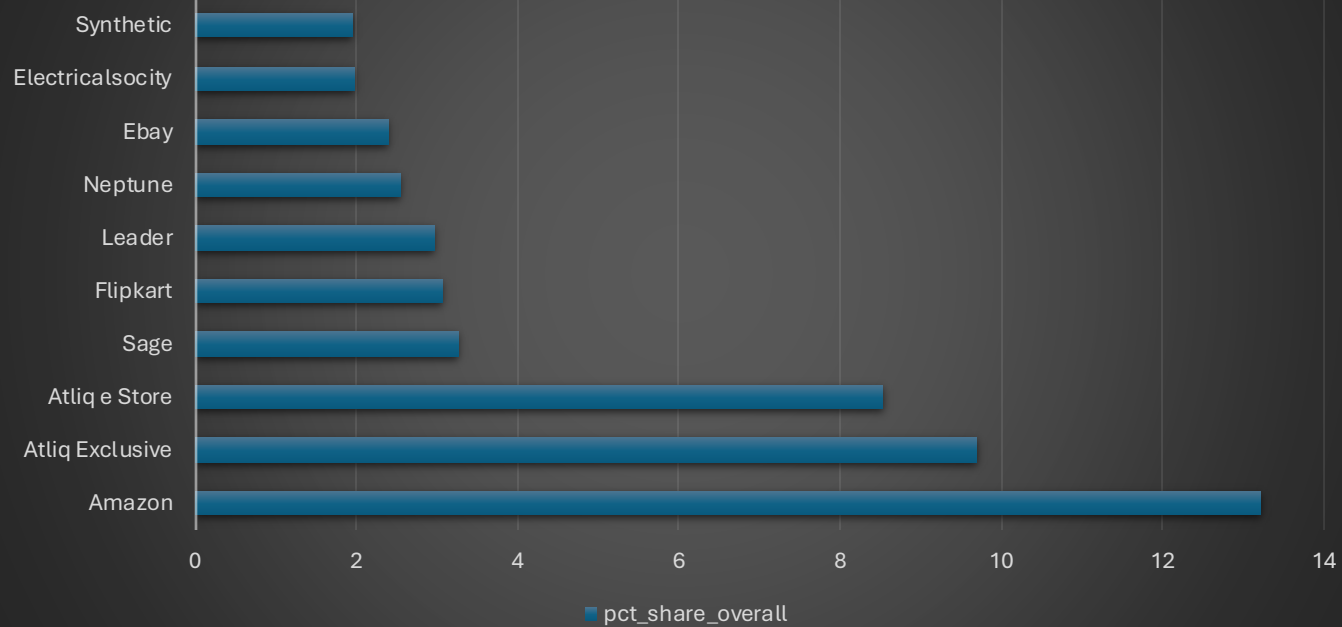
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Conclusion

- Amazon recorded the highest net sales of 109.03M in fiscal year 2021, followed by AtliQ Exclusive with 79.92M.
- India generated the highest net sales in the APAC market with 210.67M in fiscal year 2021, followed by the USA with 132.05M.
- Amazon contributed 13.23% of the total net sales among all customers in fiscal year 2021.
- Identified the top product by quantity in each division, with AQ Gamers Ms recording the highest quantity.
- In APAC region, Amazon contributed maximum net sales in 2021

Visuals

pct_share_overall



net_sales_mln

